

User Manual – Instructions for End Users

Overview

This user manual explains how to use the Brand Feedback Analyzer project as an end user. The system is a Streamlit-based application that collects Reddit posts and comments, filters Egypt-relevant brand content, predicts sentiment with a trained machine learning model, and presents recommendation and monitoring views in an interactive dashboard.

The application is designed around four main user-facing capabilities: brand data collection, sentiment analysis, recommendation ranking, and MLOps workflow tracking. It also supports Reddit OAuth credentials through Streamlit secrets or environment variables so users can access Reddit data more reliably and reduce 403 errors during fetching.

Intended Users

This manual is intended for non-developer or semi-technical end users who want to analyze public Reddit feedback about Egyptian brands using the dashboard interface and prebuilt pipeline. Typical use cases include comparing telecom or fintech companies, checking customer sentiment, and reviewing recommendation scores and pipeline status from a single interface.

System Functions

The application provides the following core functions to end users:

- Search and collect Reddit posts and comments for a selected brand, including Egyptian and Arabic aliases when available.
- Clean and preprocess collected text by removing URLs, HTML, punctuation, emojis, and Arabic diacritics before inference.
- Predict sentiment labels using a saved SVM classifier and TF-IDF vectorizer.
- Build company profiles and rank brands using sentiment distribution, engagement, and confidence-based scoring.
- Display charts, ranking tables, comparison cards, dataset statistics, evaluation metrics, and workflow monitoring panels in Streamlit.

Requirements for Use

Before using the application, the following items should be available:

Requirement	Purpose
Python environment with project dependencies installed	Needed to run Streamlit and the ML pipeline
Trained model files in models/	The app loads svm_model.pkl and vectorizer.pkl for sentiment prediction
Dataset folders in data/	Raw, labeled, and processed data are used across the pipeline and dashboard
Reddit API credentials (recommended)	OAuth support improves Reddit access and helps avoid request failures

Launching the Application

To start the user interface, open a terminal in the project directory and run the Streamlit application file. The dashboard entry point is `app/streamlit_app.py`, which initializes the visual interface and loads the analysis components and monitoring views.

Recommended steps:

1. Open the project folder in the terminal.
2. Activate the Python virtual environment used for the project.
3. Run `streamlit run app/streamlit_app.py` to launch the dashboard locally.
4. Open the local URL displayed by Streamlit in the web browser.

Configuring Reddit Access

The application can read Reddit credentials from `.streamlit/secrets.toml` or from environment variables named `REDDITCLIENTID` and `REDDITCLIENTSECRET`. If credentials are missing, the app can still attempt unauthenticated requests, but some Reddit endpoints may return 403 errors or rate limits.

A typical secrets file structure is shown below:

```
[reddit]
clientid = "YOUR_CLIENT_ID"
clientsecret = "YOUR_CLIENT_SECRET"
```

Using the Dashboard

After launch, the dashboard acts as the main end-user workspace for brand analysis, recommendation viewing, and pipeline monitoring. The interface includes analytics cards, charts, ranking tables, comparison panels, and a dedicated MLOps workflow view that summarizes model and data status.

Brand Analysis Workflow

To analyze a brand through the dashboard, the user enters or selects a brand, triggers data collection, and waits for the app to retrieve Reddit posts and comments using brand queries and aliases. The system then filters for Egypt relevance, cleans the text, predicts sentiment, and stores the processed outputs for later ranking and display.

End-user workflow:

1. Enter the target brand name in the interface.
2. Start analysis or re-analysis from the relevant dashboard button.
3. Wait while the application prepares search queries, fetches posts and comments, and builds the dataset.
4. Review the resulting sentiment outputs, charts, and recommendation tables.

Recommendation View

The recommendation system ranks companies using positive percentage, negative percentage, engagement, and confidence-derived scoring. The dashboard shows a top company, recommendation score, positive feedback percentage, negative feedback percentage, and a reason string explaining the ranking outcome.

Brand Comparison

The dashboard supports comparison cards for multiple results, showing total records, positive versus negative percentages, average score, and recommendation score per brand. This view is useful when an end user wants to compare competing brands side by side rather than reviewing one brand in isolation.

MLOps Workflow Page

The MLOps page provides a workflow tracker with stages such as data collection, data cleaning, feature engineering, model training, model evaluation, deployment, and monitoring. It also displays evaluation metrics including accuracy, precision, recall, F1 score, and loss when the labeled dataset and model artifacts are available.

Understanding the Outputs

The dashboard and pipeline produce several output types that users should understand before interpreting results.

Output	Meaning
Raw data files	Reddit posts and comments collected for the chosen brand
Processed prediction files	Cleaned records with predicted sentiment labels
Recommendation ranking	Ordered list of companies scored by sentiment, engagement, and confidence
Evaluation metrics	Model quality indicators computed from labeled evaluation data when available
MLOps snapshot	Dataset counts, feature counts, model presence, and deployment-style status summary

Sentiment Labels

The project uses three sentiment classes: positive, neutral, and negative. During training, category labels are mapped into these classes and used by a LinearSVC model trained on TF-IDF text features.

Data Processing Behavior

Text cleaning removes HTML tags, URLs, Arabic diacritics, punctuation, symbols, emojis, and numbers so the classifier receives a normalized input representation. The preprocessing logic also applies token-based normalization and lightweight stemming rules before vectorization and prediction.

Running the Pipeline Outside the Dashboard

Advanced end users can run the end-to-end pipeline from the command line using `src/run_pipeline.py`, which prompts for a brand, collects Reddit data, predicts sentiment, saves raw and processed CSV files, and prints a ranking summary if company profiles are available. The training step can also be run separately through `src/train_model.py` to regenerate the saved SVM model and TF-IDF vectorizer from the labeled dataset.

Troubleshooting

Common issues and likely causes are listed below based on the project behavior.

Issue	Likely Cause	Suggested Action
Reddit fetch returns 403	Missing or invalid OAuth credentials	Add valid Reddit credentials in <code>.streamlit/secrets.toml</code> or environment variables
No results for a brand	No Egypt-relevant Reddit records matched the brand filters	Try another brand spelling or wait for fallback global search behavior
Prediction cannot run	Model or vectorizer files are missing	Ensure <code>models/svm_model.pkl</code> and <code>models/vectorizer.pkl</code> exist
Evaluation metrics show zeros	Labeled evaluation file or model artifacts are unavailable	Verify labeled data and trained model files are present
Dashboard loads but ranking is empty	No processed company prediction files were found	Run analysis first so prediction CSV files are generated

Safe Usage Notes

This tool is designed for public sentiment exploration and comparative brand analysis rather than official financial, legal, or regulatory decision-making. Since the outputs depend on available Reddit content, query coverage, and model predictions, results should be interpreted as decision support rather than guaranteed ground truth.

File Roles for End Users

The main files relevant to end users are summarized below.

File	End-user relevance
<code>app/streamlit_app.py</code>	Main dashboard interface
<code>src/scraper.py</code>	Collects Reddit brand data
<code>src/predict_reddit.py</code>	Applies sentiment prediction to collected data
<code>src/train_model.py</code>	Trains the sentiment model
<code>src/recommendation.py</code>	Builds ranking and recommendation outputs
<code>src/run_pipeline.py</code>	Runs the full command-line pipeline

Conclusion

The Brand Feedback Analyzer gives end users a structured way to collect Reddit brand feedback, classify sentiment, compare Egyptian companies, and inspect workflow health through one Streamlit interface. When the required model files, datasets, and Reddit credentials are configured correctly, the application supports both quick brand analysis and broader monitoring of the machine learning pipeline behind the results.